HDOC Day-64

C Programming Activity: Find the Repeated Element Using Pointers and XOR

Aim

To read an integer array and find its only repeated element using a function that receives the array through a pointer. The solution uses XOR and takes one pass through the array.

C Program

```
#include <stdio.h>
#include <stdlib.h>

void duplicate_element(int *nums, int size)
{
    int result = 0;

    /* XOR every array element.
       The repeated element remains because
       all other values occur once. */
    for (int i = 0; i < size; i++)
    {
        result ^= nums[i];
    }

    /* XOR all distinct values from 1 to size-1.
       This assumes the array values are in the
       range 1 to size-1, with one value repeated. */
    for (int i = 1; i < size; i++)
    {
        result ^= i;
    }

    printf("Repeated element: %d\n", result);
}

int main(void)
{
    int size;

    printf("Enter the size of the array: ");
    if (scanf("%d", &size) != 1 || size < 2)
    {
        printf("Invalid array size.\n");
        return 1;
    }

    int *nums = malloc((size_t)size * sizeof(int));
    if (nums == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    printf("Enter %d array elements: ", size);
    for (int i = 0; i < size; i++)
    {
        if (scanf("%d", &nums[i]) != 1)
        {
            printf("Invalid input.\n");
            free(nums);
            return 1;
        }
```

```
    }

    duplicate_element(nums, size);

    free(nums);
    return 0;
}
```

Sample Input and Output

Example 1
Input size: 5
Array: 1 3 4 2 2
Output: Repeated element: 2

Example 2
Input size: 5
Array: 3 1 3 4 2
Output: Repeated element: 3

Example 3
Input size: 6
Array: 1 4 5 2 3 4
Output: Repeated element: 4

How it Works

The function duplicate_element(int *nums, int size) receives the address of the first array element. XOR cancels equal pairs (x ^ x = 0), so the duplicate is found by XORing the array values and the expected distinct values from 1 to size-1.

Note

This XOR method assumes the array contains values from 1 to size-1, with exactly one value repeated once and all other values appearing once, as in the given examples.